Carryover Copilot: Deployment Cost & Cosmos ROI Presentation Pack

Prepared from Oct 30 Business Case Notes • Strict Separation of Measured Data, Equipment Price Estimates, and Plant-Specific Inputs (GEMINI.md Rule 5)

Executive Cost Design Principle: Steps 1–3 (Pre-fill risk ranking, Post-fill ~0.7 ms color check, and Operator unsure-zone review) make 100% of operational pass/fail decisions and run on an ordinary CPU with zero GPU hardware. Cosmos is never called per vial: explanations are drafted once per recipe-change pair, reviewed once by a human, and cached. Consequently, GPU compute is a minor, occasional expense (~$5 rented for a 10-recipe line), whereas human review time and plant-specific scrap/flush rates govern the business case.

1. System Architecture & Step-by-Step Cost vs. Measured Performance

Every stage of Carryover Copilot separates real-time line execution (Steps 1–3) from offline explanation caching (Step 4). All performance metrics below are strictly measured from project evaluations (Video 5, Card 3b, and Card 9).

Step	What It Does	Hardware Needed	Data Type	Measured Result (docs/results.md)
1. Before the Fill	Risk light per recipe change & queue reorder suggestion	Any standard PC (no GPU)	[MEASURED]	Ranks changes in right order (Spearman **0.79 HIGH**, **0.82 LOW**)
2. After the Fill	Color check of every vial against its recipe's reference	One fixed camera + any PC (no GPU)	[MEASURED]	**~0.7 ms** math/vial; tall vials **8 of 8 caught (100%)**, **0 false alarms** (1 judge, Video 5)
3. Unsure Zone	Operator visually inspects borderline vials	Nothing ($0 hardware)	[MEASURED]	**2 vials of 20 (10%)** flagged for operator look in Video 5
4. Explanation (Optional)	Cosmos drafts explanation once per recipe pair; human approves & caches	GPU w/ ~17 GB VRAM only while drafting (rentable)	[MEASURED]	**1 of 8 (12.5%)** drafts fully correct; **7 of 8** had errors → human review required

2. Upfront Hardware & Setup Cost Comparison: Modern vs. Older Factory Line

Deploying Carryover Copilot on a modern automated line (such as CSI Station 3) requires zero new camera hardware and can run on an existing station PC ($0 equipment) or a dedicated small industrial PC ($600–$2,000). Retrofitting an older manual line adds only a machine-vision camera ($300–$1,500) and an even LED backlight or white backdrop ($100–$500).

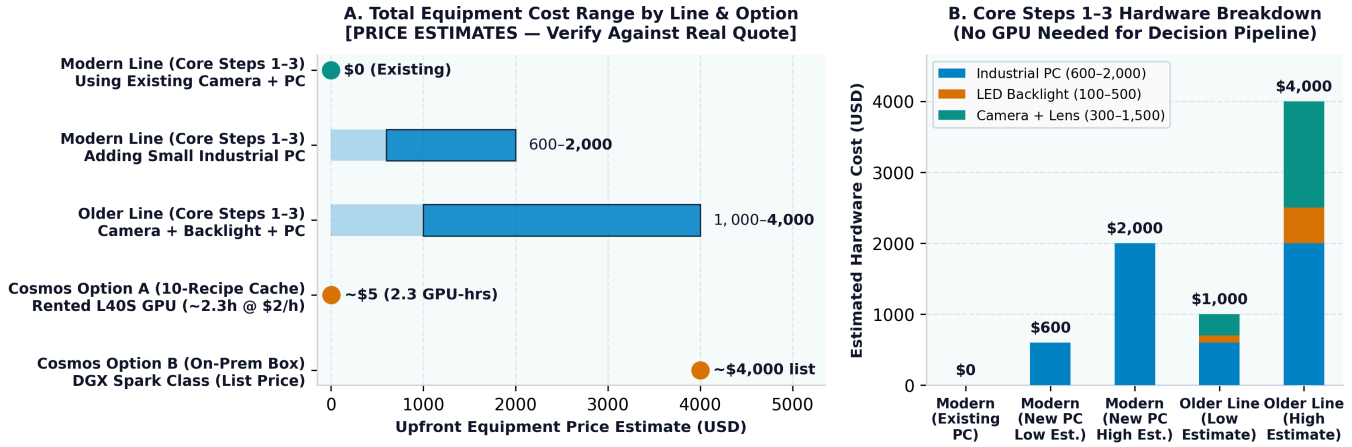


Figure 1: Upfront equipment price estimates [PRICE ESTIMATE — verify against real quote]. Core deployment (Steps 1–3) costs $0–$2,000 on a modern line and $1,000–$4,000 on an older line, with zero GPU required on the factory floor.

Item / Requirement	Cost Type	Modern Line (e.g., CSI Station 3)	Older Line (No Camera, Paper Sheets)	Est. Range
Camera + Lens + Mount	Existing / Price Est.	**$0** (reads existing `vial_fill` station camera)	Industrial machine-vision camera + lens + mount	$0 vs. $300–$1,500
Even Lighting / Backdrop	Existing / Price Est.	**$0** (existing station lighting)	LED backlight or white backdrop (mandatory: side light fooled check in twin test)	$0 vs. $100–$500
PC for Steps 1–3	Price Estimate	Existing station PC (**$0**) or small industrial PC (**$600–$2,000**)	Small industrial PC (no GPU needed)	$0–$2,000
GPU for Cosmos Drafting (Optional)	Price Estimate	Rent L40S on Brev at **~$2/hr** (~$5 total for 90 pairs), DGX Spark (**~$4,000 list**), or existing GB10 box (**$0**)	Same as modern line (cached offline approach; no line GPU)	$0 / ~$5 / ~$4,000
Reading Recipe Queue	Engineering Time	Read-only link to batch scheduler/MES, or operator types queue	Operator types the queue, or simple spreadsheet import	[Eng. Time]
Setup per Camera	Engineering Time	Place & check measurement patches once (glass-edge check flags moved camera)	Same as modern line (one-time patch placement & verification)	[Eng. Time]
Setup per Recipe	Product	**3 fills per recipe** to obtain clean reference vial (the 3rd vial)	Same as modern line (**3 fills per recipe**)	3 vials / recipe

Item / Requirement	Cost Type	Modern Line (e.g., CSI Station 3)	Older Line (No Camera, Paper Sheets)	Est. Range
PLC Changes	None	**None ($0)** — system only observes and advises	**None ($0)** — system only observes and advises	$0
Total Core Equipment (Steps 1–3)	**Price Estimate**	**$0 (existing PC) to $600–$2,000 (dedicated PC)**	**$1,000 to $4,000 total hardware**	**$0–$4,000**

3. Do Factory Lines Need Cosmos? Measured Accuracy & People vs. GPU Cost

Our measured answer is **no for deciding**, and **no for this single well-understood failure mode**. Card 3b and Card 9 evaluations demonstrate that deterministic color math (~0.7 ms/vial on CPU) catches 100% of tall bad vials (8 of 8) with 0 false alarms, whereas the best Cosmos VLM configuration caught only 5 of 8 with 3 false alarms. For explanations (Card 9), Cosmos produced only 1 of 8 fully accurate drafts while claiming `confidence: high` on all 24 responses.

Evaluation Question	Measured Evidence	Verdict	Factory Deployment Recommendation
Does Cosmos need to detect bad vials?	Card 3b: Color math caught **8/8 (0 false alarms)** at ~0.7 ms; best Cosmos caught **5/8 (3 false alarms)**	**No — Never**	**Must-Have:** Deploy camera color check + risk advisor on CPU. Never let Cosmos make pass/fail decisions.
Does Cosmos need to explain carryover on this line?	Card 9: **1/8 fully correct**; invented misrouted tubing & cleaning solution with `confidence: high` (24/24)	**Not on its own**	For a single known failure mode (previous recipe tint), a deterministic fill-in-the-blank template using previous recipe + color shift + risk light is 100% accurate at $0 GPU cost.
Where does Cosmos add real value?	Untested on our dataset (multi-cause visual anomalies)	**Optional (Cached + Reviewed)**	Use occasionally on lines with varied visual failure modes (foam, bubbles, splashing, layering, underfill, dripping nozzle) where color math alone cannot describe a novel physical defect.

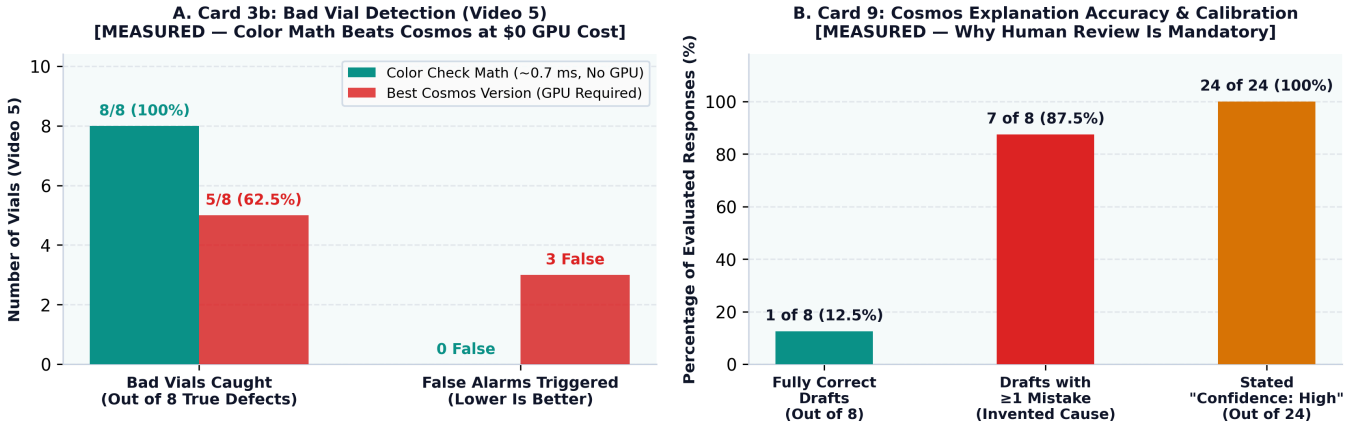


Figure 2: Measured results from Card 3b (left) and Card 9 (right) [MEASURED]. Color math outperforms Cosmos on defect detection without a GPU, and Cosmos's 87.5% error rate with 100% high-confidence claims mandates human review.

Because Cosmos is only called once per ordered recipe pair (***N* × (*N* − 1)**), a line with 10 recipes has at most **10 × 9 = 90** possible transitions. Even with 3 repeats at an illustrative 30 seconds per call, total GPU compute is only ~2.25 hours (~$5 at $2/hr). The true cost of Cosmos is human engineering review time.

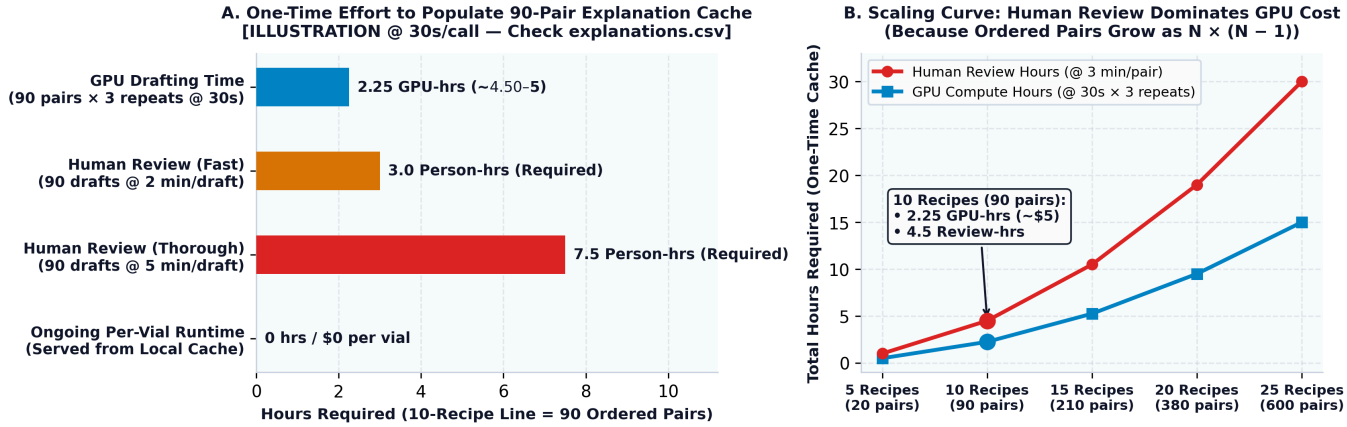


Figure 3: One-time cache population effort for a 10-recipe line (90 pairs) [ILLUSTRATION @ 30s/call — replace with measured average from `outputs/card9/explanations.csv`] and scaling curve as recipe count grows.

Cosmos Cost / Effort Component	Number Type	Formula / Calculation Basis	10-Recipe Line Value (90 Ordered Pairs)
Possible Recipe Changes	Exact Math	***N* × (*N* − 1)** ordered pairs	**90 pairs** (for ***N* = 10** recipes)
Seconds per Cosmos Call	[MEASURED / NEEDED]	Mean of `seconds` column in `outputs/card9/explanations.csv`	**[Pull from CSV]** (Illustration uses 30 s/call)
Total GPU Drafting Time	Illustration / Formula	**90 pairs × repeats × seconds/call**	**~2.25 to 2.3 GPU-hours** (at 3 repeats × 30 s)
Rented Cloud GPU Bill	[PRICE ESTIMATE]	**GPU-hours × $2/hr** (Brev L40S planning rate)	**~$4.50 to ~$5.00 total** (one-time cache build)

Cosmos Cost / Effort Component	Number Type	Formula / Calculation Basis	10-Recipe Line Value (90 Ordered Pairs)
Human Review & Approval	[YOUR NUMBER]	**90 drafts × review min/draft × labor rate**	**3.0 to 7.5 person-hours** (at 2–5 min/draft; required due to 7/8 error rate)
Ongoing Per-Vial Cost	Architecture Fact	Served from approved local cache; re-run only for new recipes	**$0 / 0 GPU-hours** during production runs

4. Operational Savings Model, Video 5 Levers & Required Plant Inputs

In Video 5, the advisor rated 8 of 9 recipe changes RED and 1 AMBER, and the color check flagged the first vial after 8 of 9 changes (all except **8 → 9**) plus the second vial in two pale recipes (vials 5 and 29). Because almost every changeover on this recipe mix is risky, skipping flushes saves little here; the primary value levers are **(1) catching off-color vials before they ship** (which the existing Teledyne check cannot do because it lacks recipe awareness) and **(2) reordering the recipe queue to eliminate RED changes**.

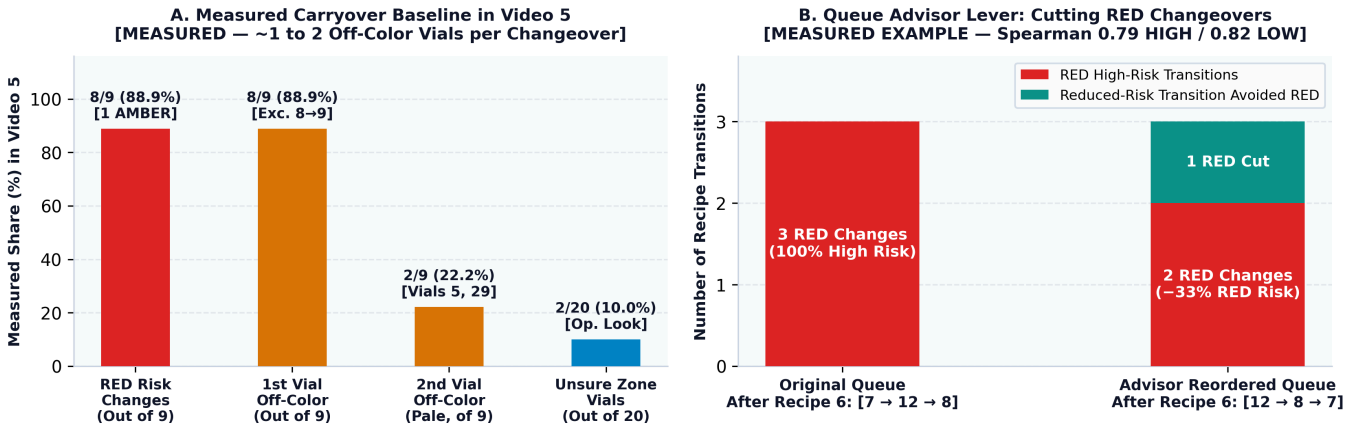


Figure 4: Measured operational baseline from Video 5 (left) and queue reordering risk reduction after Recipe 6 (right) [MEASURED].

Annual Value Formula & Plant Input Checklist (GEMINI.md Rule 5)

Per Rule 5, plant-specific financial inputs remain explicit formula variables until provided by CSI or a mentor:

Yearly Value = `(Changeovers/yr × [1 to 2 off-color vials/changeover] × [YOUR NUMBER: Cost of 1 off-color vial reaching customer or found late]) + (Flushes avoided by reorder × [YOUR NUMBER: Cost of 1 flush]) + (Operator minutes saved diagnosing × [YOUR NUMBER: Labor cost/min]) − System Cost (Section 2)`

Model Variable / Question	Category	Known / Measured Project Value	Status / Action Needed from CSI or Mentor
Off-Color Vials per Changeover	[MEASURED]	**~1, sometimes 2 vials** (8/9 1st vials + 2/9 2nd vials in Video 5)	Complete — measured directly in Video 5

Model Variable / Question	Category	Known / Measured Project Value	Status / Action Needed from CSI or Mentor
Queue Reorder RED Reduction	[MEASURED]	Cuts **3 RED changes to 2 RED** (queue `7, 12, 8` → `12, 8, 7` after Recipe 6)	Ask CSI/mentor whether queue reordering is allowed in practice & who approves
Cost of 1 Scrapped / Late-Found Vial	[YOUR NUMBER]	*Unfilled formula blank* (never guess per Rule 5)	**Ask CSI / mentor** for unit scrap cost vs. customer defect cost
Cost & Duration of 1 Line Flush	[YOUR NUMBER]	*Unfilled formula blank* (never guess per Rule 5)	**Ask CSI / mentor** for flush cost and downtime minutes
Changeovers per Shift or Year	[YOUR NUMBER]	*Unfilled formula blank* (never guess per Rule 5)	**Ask CSI / mentor** for shift changeover frequency
Measured Seconds per Cosmos Call	[MEASURED]	*Recorded in repo* (30 s used as illustration only)	**Read mean of seconds column** from `outputs/card9/explanations.csv`

5. Factory Recommendations, Cross-Industry Scope & Judge Pitch

- **1. Must-Have (Core Line):** Deploy the camera color check and risk advisor. Cheap ($0–$2,000 modern line; $1,000–$4,000 older line), fast (~0.7 ms/vial), and requires no GPU.
- **2. Optional (Explanation Drafting):** Use Cosmos only as an offline explanation-drafting tool on rented GPU (~$2/hr) or an existing GB10/DGX-class box, always gated behind human review and cached per recipe change. Worthwhile only when a line has varied visual failure modes or many recipes; unnecessary for a single known cause.
- **3. Never:** Never allow Cosmos to make the real-time pass/fail decision or display unreviewed explanations to operators.
- **Where Else This Applies:** Any production line that switches colored or flavored liquids through a shared manifold or path — including paints and coatings, printing inks, beverages, cosmetics, liquid pharmaceuticals, and chemical dosing.
- **How to Frame It to the Judges:** We used Physical AI where it earns its place and measured honestly where it does not. Isaac Sim (digital twin) let us test asymmetric lighting changes we could not stage on the physical line; Cosmos is tested, scored, and gated behind human review; and real-time quality decisions remain with fast, explainable color math.